

Shubhangi Jeve

B.Tech in Artificial Intelligence & Data Science
Terna College of Engineering, Dharashiv | CGPA: 8.5
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CAREER OBJECTIVE

AI Engineer with hands-on production experience building end-to-end LLM-powered systems, RAG pipelines, and full-stack AI applications. Proficient in Python, FastAPI, vector databases, and LLM integration using Groq, Gemini, and Hugging Face. Seeking an AI Engineer role to architect and deploy scalable, enterprise-grade intelligent systems.

EXPERIENCE

- **AI Engineer Intern**
COGNITBOTZ

November 2025 – Present
Hyderabad, Telangana
- **Project: LegalAID** — Enterprise-grade, AI-powered Legal Research and Litigation Assistance Platform for Indian court case data, combining Retrieval-Augmented Generation (RAG), semantic search, and LLM-driven drafting into a production-ready system.
 - **RAG Pipeline Architecture:** Designed and implemented a multi-stage hybrid retrieval pipeline combining dense vector search (pgvector with BAAI/bge-m3 embeddings) and PostgreSQL full-text search for lexical recall. Applied parent-child chunk retrieval with entity-aware chunking strategies tailored per legal document type including case summaries, hearing history, orders, and procedural timelines to maximize citation-grounded answer quality.
 - **LLM Integration and Hallucination Control:** Integrated Groq-hosted LLMs through a provider-abstracted service layer supporting model swapping at configuration time. Enforced a strict abstention policy where the system returns "Insufficient evidence" when retrieval confidence falls below a defined threshold. Validated every generated answer against cited source chunks to minimize hallucination in legal responses.
 - **Data Ingestion and Normalization Engine:** Built a robust ingestion pipeline that parses yearly Indian court JSON dumps containing 20,000 to 75,000 cases spanning 25 years. The pipeline normalizes 30+ canonical fields including parties, acts, bench type, coram, and hearing history, deduplicates records, generates semantic embeddings, and writes structured data with vector indexes in a single resumable run.
 - **Backend API Development (FastAPI and SQLAlchemy 2.0):** Developed a fully asynchronous REST API with complete Pydantic v2 schema validation, exposing endpoints for case search, RAG-based question answering, AI-assisted legal document drafting (notices, replies, case briefs, legal memos), litigation comparison, and a dashboard analytics module.
 - **Frontend Development (Next.js 14 and TypeScript):** Built the complete frontend in Next.js 14 with TypeScript, covering a natural-language search workspace, faceted case explorer, structured case detail pages with hearing timelines, a legal drafting workspace, side-by-side case comparison view, and an admin console for ingestion pipeline management.
 - **Vector Database and Hybrid Search:** Configured PostgreSQL with pgvector as the unified store for both structured metadata queries and approximate nearest-neighbor (ANN) vector retrieval, achieving sub-4-second search latency on the full case corpus while eliminating the overhead of a separate vector database.
 - **Containerized Deployment with Docker:** Packaged the entire system using Docker and Docker Compose, with the FastAPI backend running in a lightweight Python 3.11 image and the vector-enabled database on PostgreSQL 16 with pgvector. Implemented health-check dependency chains ensuring correct startup order and persistent named volumes for data durability.

- **Project: MeetOps (May 2026 – Present)** — Enterprise-grade AI-powered Meeting Copilot and Project Intelligence Platform integrating Microsoft Teams via ephemeral Dockerized Playwright bots, a RAG-driven knowledge engine, and an agentic LLM layer that transforms live meeting context into structured, actionable project intelligence.
- **RAG Pipeline and Vector Intelligence:** Designed a semantic RAG pipeline over meeting transcripts, BRDs, and project artifacts using OpenAI embeddings stored in pgvector. Applied speaker-turn and topical-boundary chunking (over fixed character splits) to preserve semantic coherence, significantly improving top-K retrieval precision for time-sensitive project queries.
- **Hallucination Control and Confidence Gating:** Enforced a strict abstention guardrail where the LLM is instructed to respond “Insufficient meeting context” when cosine similarity of retrieved chunks falls below 0.75, preventing fabricated technical details in high-stakes project conversations.
- **Agentic Tool-Calling and LLM Orchestration:** Implemented an agentic workflow using OpenAI function-calling to trigger structured actions mid-conversation, including Jira ticket creation, rolling meeting summaries, and live SAP GRC metrics retrieval, transforming the copilot from reactive Q&A into an autonomous project action layer.
- **Ephemeral Bot Orchestration (Docker + Playwright):** Engineered a Disposable Workspace bot pattern where per-meeting Playwright instances are dynamically provisioned and reaped via Docker with TTL watchdogs and health checks, ensuring zero state contamination across client sessions and efficient compute utilization.
- **Low-Latency Streaming and Caching:** Implemented a multi-tier Redis caching layer that short-circuits the full embedding and LLM generation cycle on repeated or semantically identical queries, and streamed LLM output directly to the React frontend via Server-Sent Events (SSE) to minimize perceived latency during live meetings.

• Artificial Intelligence Intern

February 2025 – April 2025

Infosys Springboard

Remote

- Built a Healthcare RAG Chatbot on a Wikipedia dataset using FAISS vector search, a Flask backend, and a React with Tailwind CSS frontend, with secure API endpoints designed for medically grounded responses.
- Implemented a real-time query processing pipeline with source verification and an interactive UI supporting multi-turn question answering sessions.

• Data Science Intern

January 2025 – June 2025

Adhyayan IT

Remote

- Built a Text-to-SQL application using LangChain and LLMs (Google Gemini, Groq) that converts natural language queries to executable SQL, with RAGAS evaluation achieving 100% context precision and high helpfulness scores.
- Developed an interactive Streamlit UI with secure MySQL connectivity, enabling non-technical users to query databases efficiently with reliable, context-aware responses.

• Data Analyst Intern

October 2023 – January 2024

Cyber Police Station

Dharashiv, Maharashtra

- Managed 500+ user credentials for the National Cyber Crime Reporting Portal and organized 1,000+ complaint records, improving case resolution efficiency by 30% and accelerating case progression by 20%.
- Communicated with victims to provide timely status updates, enhancing overall satisfaction by 15%.

PERSONAL PROJECTS

• RAG Document QA System

Production-grade AI-powered question answering on PDF and TXT documents using FastAPI and React

- Engineered a full-stack RAG system with a FastAPI backend and a React (TypeScript, Tailwind CSS) frontend, integrating document chunking, sentence-transformers embeddings, and ChromaDB vector search with source verification for accurate and traceable answers.
- Built asynchronous RESTful APIs and deployed the complete stack using Docker, with an interactive dashboard for monitoring ingestion KPIs and retrieval quality metrics in real time.

• Insurance Claim Automation System

AI-powered insurance claim processing using Google Gemini AI, LangChain, and Flask

- Developed an AI-powered insurance claim automation system using Google Gemini AI to process and validate medical claims from PDF documents, extracting key medical and billing information using natural language processing.
- Built a Flask web application with LangChain integration for intelligent claim assessment and decision-making, with robust error handling and data validation pipelines to ensure accurate processing and reduce manual review time.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, FastAPI, Flask, Next.js 14 (TypeScript), Streamlit

AI and Machine Learning: Regression, Classification, Clustering, Feature Engineering, ANN, CNN, Fine-tuning, Prompt Engineering

Generative AI and LLMs: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Groq API, Google Gemini, Ollama, Open-source LLMs

Embeddings and Vector Search: sentence-transformers (BGE-M3, multilingual-e5), FAISS, ChromaDB, pgvector, Hybrid Semantic and Lexical Search

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow/Keras, PyTorch, NLTK, spaCy, Hugging Face Transformers, Pydantic v2, SQLAlchemy 2.0

Databases: PostgreSQL with pgvector, MySQL, ChromaDB, FAISS Vector Store, SQLite

DevOps and Tools: Docker, Docker Compose, Git, GitHub, VS Code, Jupyter Notebook, N8N, Observability Tools

EDUCATION

• Bachelor of Technology in Artificial Intelligence and Data Science

2021 – 2025

Terna Public Charitable Trust's College of Engineering, Dharashiv

CGPA: 8.5